Blood Cell Detection and Leukemia Screening with Area Matched Evidence Regularization

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***Abstract* - Automated blood image analysis must distinguish cell localization from disease screening and examine which image regions support its predictions. We propose area matched evidence regularization for acute lymphoblastic leukemia (ALL) screening within a blood cell analysis system. During training, a ranking loss compares category support after erasing an automatic stain proposal with support after erasing equally many context pixels. The classifier uses image labels and requires one feature extraction pass at inference. On 643 previously evaluated images, it achieves 98.0% accuracy across four categories, against 98.8% for training on original images. The separate cell detector achieves mean average precision 0.910 at an overlap threshold of 0.50. Controlled comparisons show that the ranking loss does not improve clean accuracy and reduces robustness to global colour changes. The contribution is a training formulation for ALL and a controlled test of its predictive effect; independent patient evaluation remains necessary.**

***Index Terms* - blood cell detection, acute lymphoblastic leukemia, evidence guided learning, image classification, source grouping**

I. Introduction

Microscopic blood images support the localization of red blood cells (RBCs), white blood cells (WBCs) and platelets, as well as analysis of abnormal cell patterns. Cell detection, cell subtype classification and disease screening are distinct tasks with different annotation requirements [1]-[3]. A combined analysis system must therefore associate each output with the labels available for that task.

This study addresses blood cell detection and acute lymphoblastic leukemia screening. The Blood Cell Count and Detection (BCCD) collection supplies detection boxes, Blood Cell Images supplies normal WBC categories, and the ALL collection supplies benign and malignant labels for complete images [4]-[6]. The latter images contain multiple cells. A field label cannot establish which detected cell is malignant. We therefore use separate detection and screening branches, reporting cell counts alongside an ALL assessment of the image.

High image accuracy alone does not establish dependable biological evidence. Related source images can inflate performance across data partitions, and acquisition or background cues can provide shortcuts [7]-[9]. Simply removing the background can also discard useful information [10], [11]. This motivates a narrower question: can training encourage greater dependence on stained regions than on an equally large region of surrounding context, without requiring expert region annotations?

We contribute area matched evidence regularization for ALL screening. The formulation combines automatic stain proposals, context controls of equal area and a supervised class margin ranking loss. It operates on image labels without expert region annotations and adds no extra feature extraction pass at inference. Five controlled training variants separate augmentation, ranking and area effects, while grouped source auditing and fixed perturbations support a reproducible comparison. Together, the formulation and evaluation test whether this evidence preference improves screening within a system that also detects and counts blood cells.

II. Related Work

Convolutional neural networks (CNNs) have been used for blood cell recognition and ALL image classification [2], [3]. Residual networks illustrate the benefits of learned visual representations, while You Only Look Once (YOLO) models provide compact object detection and classification implementations [12]-[14]. The present work retains a YOLOv8 nano detector and uses the classification variant as a frozen feature extractor.

Evidence guided learning provides the methodological background. Ross *et al.* constrain explanations through input gradients, while Chang *et al.* generate object and background interventions to reduce spurious associations [15], [16]. Askari Farsangi *et al.* describe an ALL pipeline sensitive to biomarkers and a learning approach that combines several images [11]. More recently, Evidence-RL compares evidence removal with matched nonevidence regions during reinforcement learning of models that combine vision and language [17].

TABLE I
Closest work and the specific extension tested

Work	Relationship to this study
Ross *et al.* [15]	Explanation constraints motivate evidence training; our penalty compares image erasures.
Chang *et al.* [16]	Context augmentation is an explicit control; ranking must add value beyond it.
Askari Farsangi *et al.* [11]	ALL screening guided by cell regions precedes this work; image labels here support field classification.
Huang *et al.* [17]	Matched evidence comparison precedes this work; we test a supervised ALL margin formulation.
Abdallat *et al.* [10]	Original and segmented ALL views motivate testing information loss instead of assuming benefit.

Our adaptation uses a direct margin loss for four categories with stain proposals independent of the class label in multicell ALL fields. The question is whether this supervised formulation adds value beyond context augmentation alone. Table I identifies the closest precedents and the extension evaluated here. The methodological contribution is specific to this formulation and controlled experiment; evidence guided learning and comparison of matched regions are established principles.

III. Datasets and Source Audit

A. Cell sources and leakage correction

BCCD contains 364 fields: 205 training, 87 validation and 72 test images. The test set contains 805 RBCs, 71 WBCs and 69 platelets. The legacy WBC split grouped augmented filenames by a random suffix rather than the original image index. Keras source code documents this naming pattern [18]. Recovering the source index showed that all 2,476 legacy validation images had augmentation relatives in training, even though their file hashes differed.

We rebuilt WBC classification from 339 original fields with one annotated WBC and one supported subtype. The partitions contain 191 training, 85 nominal validation and 63 test sources. Perceptual screening and affine registration using scale invariant feature transform (SIFT) descriptors found a related training and validation field [19]. Excluding that validation image leaves 84 eligible fields. The new WBC classifiers use the fixed final training epoch, not a checkpoint selected using the nominal validation scores. The inherited detector's historical checkpoint selection did include this related validation scene; this limitation is retained.

B. ALL images and source grouping

The local ALL collection contains 3,256 original images at 224×224 pixels. The categories are Benign, Early, Pre and Pro; the malignant categories denote early pre-B, pre-B and pro-B ALL, not increasing severity [3], [6]. Deduplication of identical decoded pixels removes 40 copies, leaving 3,216 images.

To group observable image relatives, we compute perceptual hashes of 64 bits under eight rotations and reflections.

TABLE II
ALL images after deduplication and grouping

Category	Training	Validation	Test
Benign	305	103	96
Early	587	200	198
Pre	574	194	195
Pro	462	148	154
Images	1928	645	643
Image groups	1888	630	629

Distances of four bits or less are grouped conservatively. Distances from five to ten additionally require affine SIFT registration, at least 12 inliers, an inlier fraction of 0.60 and image coverage of 10%. The descriptor ratio is 0.75 and registration tolerance is three pixels. These fixed screening settings identify candidate source relationships; they do not establish patient identity. Seventy accepted links produce 3,147 groups, with no conflicting category labels. Five stratified group folds with seed 2026 assign one fold to testing, one to validation and three to training.

C. Evaluation scope

The test partition of 643 images was evaluated in an earlier study of original and stain views before the present training idea was developed. We preserve that partition and disclose its reuse. All new fitting, regularization selection and calibration use only training and validation images, but this does not make the test partition newly untouched. The present extension is exploratory. Patient identifiers are unavailable, grouping is not exhaustive, and no external laboratory or prospective patient cohort is evaluated. This distinction follows reporting guidance on leakage and medical imaging studies [8], [20].

IV. Methods

Fig. 1 summarizes the training and inference branches.

A. Blood cell detection

The cell branch uses the supplied YOLOv8 nano detector with input size 640, score threshold 0.25 and nonmaximum suppression overlap 0.70. These inherited operating settings are fixed to preserve the detection reference while studying the ALL loss; they are not optimized for ALL images. Each detection receives one object record, so a WBC subtype never becomes an extra counted cell. Descriptive measurements use automatic image regions, without a claim of expert cell or nucleus segmentation accuracy.

The optional normal WBC classifier retains ten pixels around each crop to avoid a tight boundary cut. Three seeds, 42, 43 and 44, share ImageNet initialization and an AdamW schedule of 60 epochs [21], [22]. Combined leukemia mode withholds normal WBC subtype assignments. A qualitative inspection of training images showed missed prominent cells when the BCCD detector was applied to ALL fields. Without validated ALL boxes, we use the automatic stain proposals below as weak training regions.

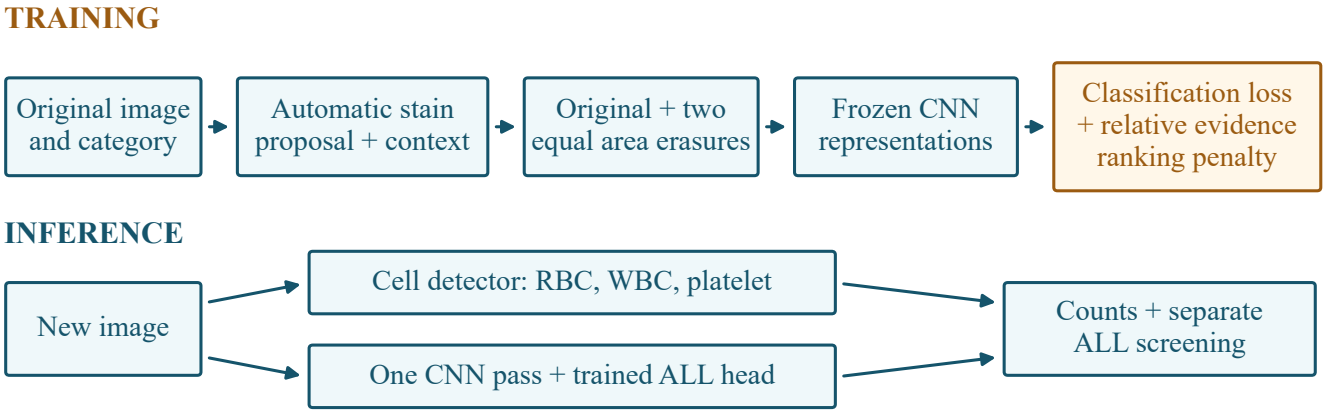


Fig. 1. Training and inference have different roles. Region interventions shape the ALL classifier during fitting. At inference, the original image supports cell detection and a separate ALL assessment. Detector boxes do not provide leukemia labels.

B. Stain and equal area context proposals

For image x, Otsu thresholding on the saturation channel of hue, saturation and value (HSV) colour space supplies an initial region [23]. This adaptive threshold avoids fixing one saturation cutoff across images. We close the region with a 3×3 kernel and dilate it with a 5×5 ellipse to reduce small gaps and preserve nearby structure. These are fixed engineering choices, not optimized clinical boundaries. If the region exceeds half the image, the most saturated half of the pixels is retained. This cap leaves sufficient complementary pixels for an equal area comparison.

Let M denote this binary stain proposal. A context mask C selects $|M|$ pixels outside M, ordered by decreasing distance to M with stable raster tie breaks. Thus

$$M \cap C = \varnothing, \qquad |M| = |C|. \tag{1}$$

Both masks are class independent. Erasure fills masked pixels with grey value 127: x_M hides the stain proposal, whereas x_C hides matched context and retains the stain proposal. Neither view isolates all diagnostic information. A common fill value and pixel count control two intervention differences, without matching shape, texture or diagnostic content. Fig. 2 shows a training example. If M is empty, its ranking term is disabled. Neither mask is an expert biological segmentation.

C. Area matched evidence regularization

The ImageNet pretrained YOLOv8 nano classifier supplies a frozen feature vector of 1,280 dimensions $\phi(x)$. Standardization uses original training features only. A linear head produces four logits $z(x) = W\phi(x) + b$. Freezing the representation isolates the proposed training loss from changes in backbone capacity and permits central processing unit (CPU) reproduction.

For true category y, define its margin against the remaining categories:

$$m_y(x) = z_y(x) - \log \sum_{k \neq y} \exp z_k(x). \tag{2}$$

Area matched evidence regularization (AMER) uses the penalty

$$r_y(x) = \log\left(1 + \exp\left[1 - m_y(x_C) + m_y(x_M)\right]\right). \tag{3}$$

Minimizing it favours greater support for the true category when stained regions remain visible than when they are erased. The unit margin fixes the scale of this preference; its strength is selected through the coefficient λ. The complete objective is

$$\mathcal{L} = \tfrac{1}{2}\langle \ell_y(x) + \ell_y(x_C)\rangle_w + \lambda \langle r_y(x)\rangle_{w,V} + \tfrac{\alpha}{2}\|W\|_F^2, \tag{4}$$

where ℓ_y is cross entropy and $\langle\cdot\rangle_w$ gives equal total weight to each class. The ranking mean uses nonempty proposals V and is zero if V is empty. The two classification terms retain the source label on original and context erased images; averaging preserves the total supervised weight per source. The margin already compares the true logit with all rival logits, without a separate rival class penalty. Stain erased images receive no benign label because erasure cannot establish absence of disease. The loss specifies a preference, not a causal disease model.

D. Controls, fitting and inference

Five families use the same representation, source split and fitting routine: classification of original images; matched context augmentation without ranking; full context augmentation without ranking; AMER; and AMER with the entire complement erased instead of equal area context. The last family tests area matching while retaining the ranking objective. All controls are fitted afresh; the earlier experimental head is not substituted for the new control trained on original images.

For each family, α ranges over 0.0001, 0.001 and 0.01. Ranking families additionally use $\lambda \in \{0.1, 1.0\}$. These small logarithmic grids compare regularization strengths within a fixed fitting budget. We fit the heads deterministically from zero using the limited memory Broyden, Fletcher, Goldfarb and Shanno optimizer with all training samples and a budget

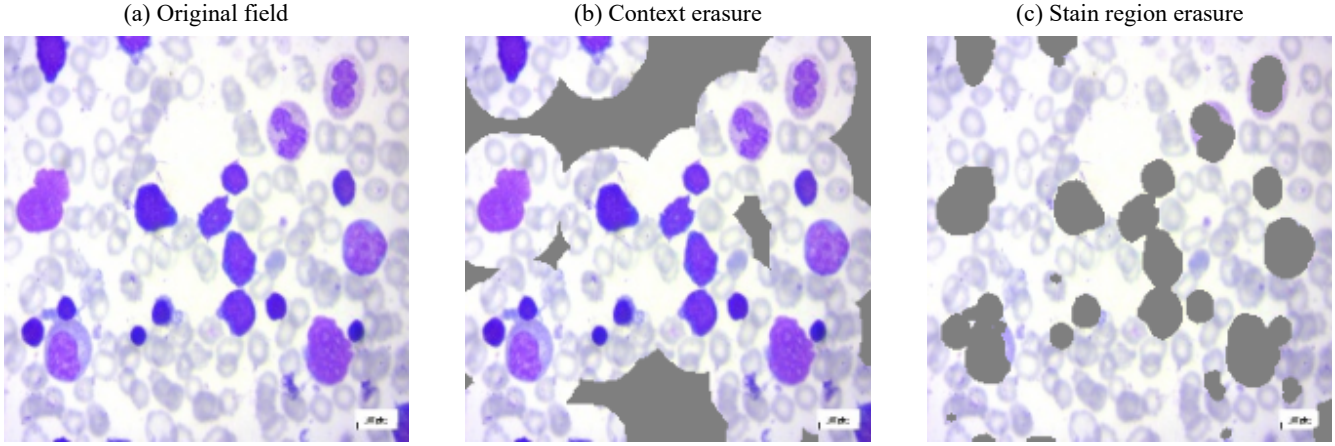


Fig. 2. Training interventions from one source image. The stained region remains visible in (b) and is hidden in (c). The two erasures affect the same number of pixels. These automatic regions provide a training hypothesis, not labelled malignant cells or validated nuclear boundaries.

of 2,000 iterations. Selection uses validation macro F1, then negative log likelihood (NLL), then stronger regularization; nonconverged candidates are ineligible. Macro F1 averages the harmonic mean of precision and recall equally over the categories. One temperature fitted on validation data calibrates the selected logits [24]. Scikit-learn supplies metric implementations [25].

All family choices and artifact hashes are saved before new test prediction. The accompanying project records source groups, split assignments, image and model hashes, and selection results; fixed code regenerates masks from source images. Inference requires only $\phi(x)$ and the selected head, without intervention views. Binary ALL support sums the three malignant probabilities. The application's review threshold is selected on validation data and is not the proposed contribution.

E. Measurements and fixed perturbations

Detection average precision (AP) summarizes precision over recall; mean AP (mAP) averages over classes. We evaluate intersection over union (IoU) 0.50 and the range 0.50 to 0.95 [26], [27]. IoU is shared box area divided by combined box area. ALL evaluation reports accuracy across four categories, macro F1, Matthews correlation coefficient (MCC), binary sensitivity and specificity [28], [29]. Sensitivity measures correctly flagged ALL images; specificity measures correctly labelled benign images.

The fixed conditions are original images, Gaussian blur with a 5×5 kernel and standard deviation one, and global blue, green and red gains of $(0.90, 1, 1.10)$. A fourth condition applies gains $(0.75, 1, 1.25)$ only outside the original stain proposal, preserving proposed evidence pixels exactly. Intensities are clipped to the valid range. The blur probes loss of detail; the colour gains probe global colour variation and sensitivity to context. These engineering stress settings do not represent calibrated laboratory variation. No parameter is retuned on them. Paired differences use 2,000 bootstrap resamples of source groups, retaining all members of each sampled group. These intervals quantify uncertainty across image groups and do not resolve patient dependence or test reuse.

TABLE III
Cell detection on 72 test fields

Category	Objects	AP at 0.50	AP at 0.50 to 0.95
RBC	805	0.864	0.616
WBC	71	0.962	0.770
Platelet	69	0.903	0.476
All classes	945	0.910	0.621

V. Results

A. Cell analysis

The detector achieves mAP 0.910 at IoU 0.50 and 0.621 across the stricter range on 72 BCCD test fields. Values for each category are shown in Table III. The primary normal WBC crop model correctly classifies 55 of 63 sources, or 87.3%, with a 95% Wilson interval of 76.9% to 93.4% [30]. The complete detector and subtype cascade correctly labels 54 true WBCs, with three unmatched predictions. These results concern the cell source dataset and do not measure combined clinical accuracy on ALL fields.

B. ALL training comparisons

All 21 fitting candidates converged under the final optimization budget. Validation selected $\alpha = 0.0001$ and $\lambda = 0.1$ for AMER. Table IV reports every family under every fixed condition. On original images, AMER achieves 98.0% accuracy and macro F1 0.977, compared with 98.8% for the head trained on original images and 98.6% for matched augmentation. Its binary counts are 540 true positives, 7 false negatives, 94 true negatives and 2 false positives. Sensitivity is 98.7% and specificity is 97.9% (Table V). The recorded NLL after temperature scaling is 0.065, against 0.048 for original training; lower values indicate better probability scores. Fig. 3

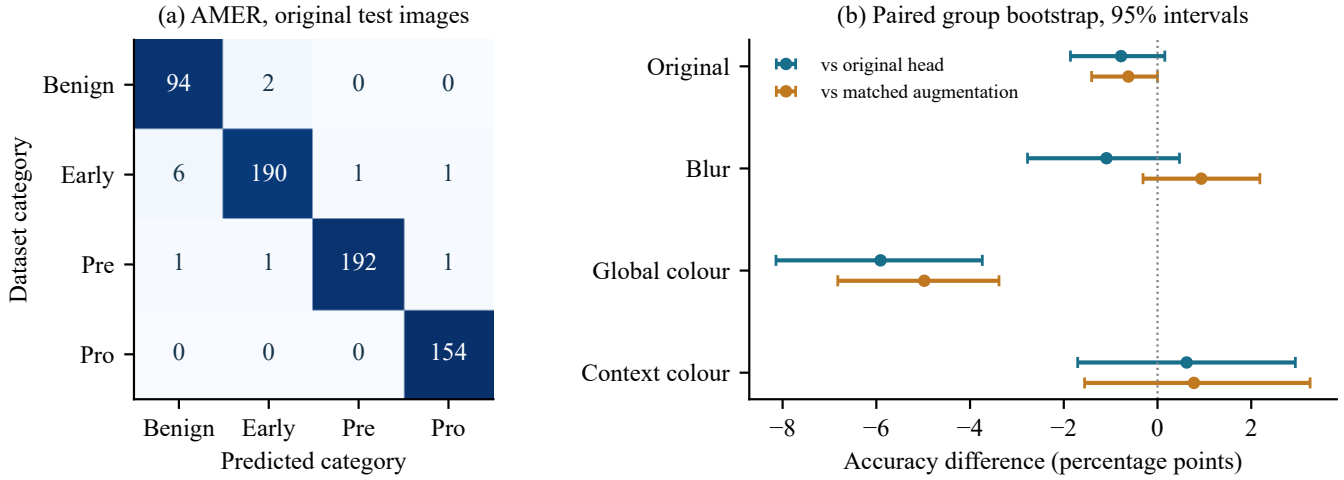


Fig. 3. Recorded results, with all 643 test images retained. The confusion matrix shows absolute counts. Difference intervals compare the proposed model with the original image and ordinary augmentation controls; the zero line makes uncertain or negative differences visible. These are exploratory comparisons of image groups.

TABLE IV
Accuracy across four categories under fixed conditions (%)

Training	Original	Blur	Global	Context
Original	98.8	90.8	92.7	73.6
Matched augmentation	98.6	88.8	91.8	73.4
Full context augmentation	98.1	93.3	92.2	78.2
AMER	98.0	89.7	86.8	74.2
Unmatched AMER	97.7	93.2	89.7	75.0

Global and Context denote global colour and changes to context colour, respectively.

TABLE VI
AMER accuracy differences and 95% group intervals

Condition	vs original	vs matched aug.
Original	-0.8 [-1.9, +0.2]	-0.6 [-1.4, +0.0]
Blur	-1.1 [-2.8, +0.5]	+0.9 [-0.3, +2.2]
Global colour	-5.9 [-8.1, -3.7]	-5.0 [-6.8, -3.4]
Context colour	+0.6 [-1.7, +2.9]	+0.8 [-1.6, +3.3]

Values are percentage points. Intervals crossing zero do not establish a difference under this resampling analysis.

locates eight of AMER's 13 category errors in Early images. All results retain every test image, before review rejection.

C. *Effects of ranking and area matching*

AMER changes accuracy on original images from 98.8% to 98.0%. Its 0.6 percentage point gain under context colour changes has a paired interval crossing zero. Against matched augmentation, the corresponding gain is 0.8 points and is also uncertain. Under global colour changes, AMER loses 5.9 points relative to training on original images, with the entire interval below zero (Table VI and Fig. 3). Thus the observed context gain does not establish greater robustness overall.

Full context augmentation reaches 93.3% under blur and 78.2% under context colour changes, whereas AMER reaches 89.7% and 74.2%. Adding ranking to full context augmentation lowers these values to 93.2% and 75.0%. Area matching also has no consistent advantage across the conditions. These controls show why an apparent improvement under one transformation is insufficient to validate the evidence assumption. No model family is selected retrospectively from the test outcomes.

TABLE V
ALL metrics on original test images

Training	F1	MCC	NLL	Sens. (%)	Spec. (%)
Original	0.986	0.983	0.048	99.5	99.0
Matched augmentation	0.984	0.981	0.054	99.1	99.0
Full context augmentation	0.979	0.975	0.072	98.9	97.9
AMER	0.977	0.973	0.065	98.7	97.9
Unmatched AMER	0.974	0.968	0.092	98.9	96.9

F1 is the macro average. NLL uses temperature scaled probabilities. Sensitivity and specificity concern the binary ALL decision.

VI. Discussion and Limitations

Matching erased area did not make automatic stain proposals an effective training target in this experiment. The supervised margin formulation turns this assumption into a trainable objective, building on explanation constraints and object and context interventions [15]-[17]. The comparisons measure predictive effects. We did not measure spatial attribution or whether support shifted toward the stain proposal.

Equal area controls pixel count, not geometry or visual impact. Saturation proposals can include RBCs or miss weak staining, and grey erasure can introduce artificial edges. The results do not isolate which factor caused the colour sensitivity. We tested one stain rule and fill value, without random context masks, alternative area caps, inpainting or stain normalization. Stronger region proposals remain untested.

The frozen backbone isolates changes to the classifier but cannot adapt its spatial features. No backbone fine tuning, region aggregation baseline or contrastive objective was evaluated. The small parameter grid does not establish that AMER fails under other settings. NLL reports probability performance after scaling; expected calibration error, Brier scores and a comparison before and after calibration were not evaluated.

The source audit reduces observable duplication but cannot recover missing patient identifiers or eliminate shared acquisition cues. The test partition was previously examined, so the extension needs independent confirmation. The cell detector is inherited, with a known historical validation overlap; the normal WBC test set includes only three monocytes and four lymphocytes. The available datasets do not jointly label cell boxes and ALL categories, so combined accuracy is not established. Larger annotated cell resources, including Raabin WBC, can support better region models, but are not substitutes for an external ALL evaluation [1], [20], [31].

VII. Conclusion

We presented a blood cell analysis system and an evidence regularizer for ALL based on equal area stain and context erasures. Its supervised margin loss requires no expert region labels or additional inference pass. The controlled comparison establishes a clear limit: the proposed ranking term did not outperform training on original images overall, despite 98.0% clean accuracy. This result supports testing evidence constraints against augmentation controls rather than assuming that region guidance improves screening. External data from separate patients and validated region annotations are the next requirements for assessing generalization.

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